

Date	15 February 2025	Team ID	6ab61594fa5e5219dc387253
Project Name	Voice Patient Diary & Clinical Integration	Maximum Marks	4 Marks

PROJECT DESIGN PHASE | ARCHITECTURE BLUEPRINT

Solution Architecture

Overview: Solution architecture is a comprehensive multi-tier framework that bridges the gap between complex business requirements and robust technology implementations. It establishes scalable design patterns, security protocols, and operational workflows across integrated cloud services.

Core Objectives & Goals

- **Technical Alignment:** Identify and deploy optimal cloud and software solutions to address core clinical and data ingestion business problems.
- **Stakeholder Communication:** Formally describe the structure, behavior, and data flow characteristics of the application for developers and stakeholders.
- **Feature & Phase Definition:** Define system features, development phases, and technical requirements for systematic project delivery.
- **Operational Specifications:** Provide precise technical specifications according to which the solution is defined, managed, and validated.

Solution Architecture Diagram

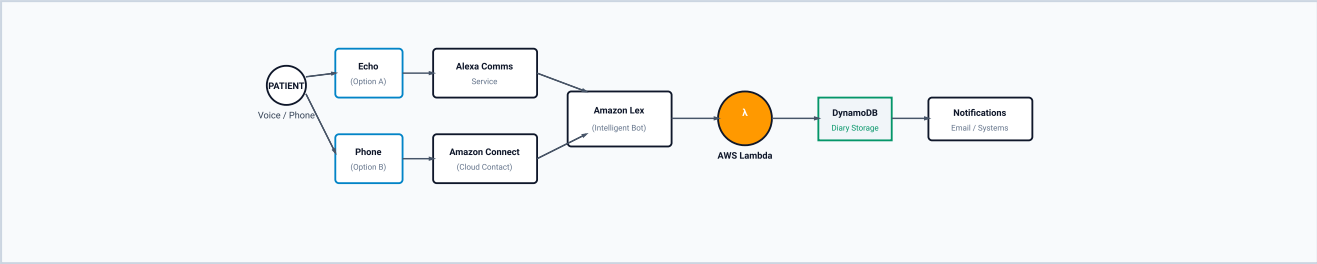


Figure 1: Architecture and data flow of the voice patient diary sample application (AWS Cloud Infrastructure)

Architecture Highlights: The solution integrates conversational voice interfaces (Amazon Alexa / Connect) with natural language understanding (Amazon Lex). Intent slots (e.g., medication logging time) are securely processed via serverless AWS Lambda functions and persisted in Amazon DynamoDB with robust encryption keys managed through AWS KMS.